Setup for Vat Photopolymerization 323

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Class Outline

Vat Photopolymerization
Software
Resin Delivery System
Machine Components
Vat Calibration
Vat Photopolymerization Component and System Review
Build Orientation
Build Surface Setup
Laser Calibration
Setup Review

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Vat Photopolymerization

Vat photopolymerization is an additive manufacturing (AM) technique that uses concentrated ultraviolet (UV) light in the form of a laser to cure, or harden, liquid photoreactive resins, or photopolymers, layer-by-layer to form a solid part. Typically, vat photopolymerization machines include both consumer-level desktop-sized machines and larger, industrial-level machines for end-use part production.

Most industrial-level vat photopolymerization machines have similar setup procedures, which include both automatic and manual tasks. An effective setup strategy creates the foundation for repeatable, consistent outcomes for each production run using industrial vat photopolymerization machines. Improper setup can create unwanted variances in components and decreased repeatability, which can also lead to increased scrap rates and the loss of time, resources, and energy.

In vat photopolymerization setup, an operator must actively employ software and understand machine settings, parameters, and vat conditions. Altogether, setup processes for vat photopolymerization can be divided into three general areas: software, the resin delivery system, and the build chamber.

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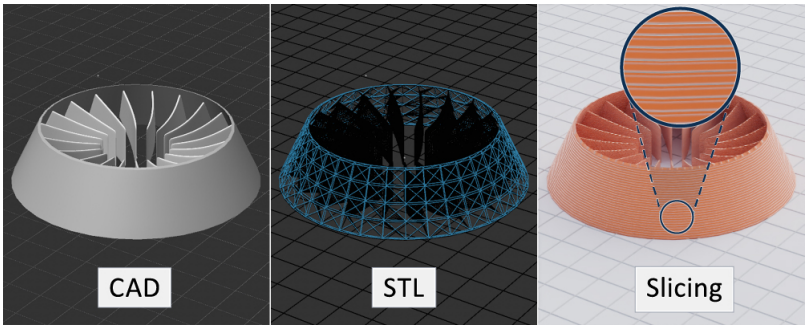
Software



Similar to other AM methods, operators using vat photopolymerization machines work with computer-aided design (CAD) software to create a three-dimensional (3D) design file of a part. Because vat photopolymerization machines cannot typically read CAD files, it is necessary to convert the 3D CAD file into an STL file format, which uses a sequence of triangles to replicate the 3D model's external surface.

Vat photopolymerization machines typically have their own proprietary software which performs the following tasks: creates support structures for the STL model, translates the STL model and supports into slices or layers, and generates G code for the layers to control the functions and motions of the machine hardware.

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Resin Delivery System

All vat photopolymerization systems include a resin tank, or vat, which all subsystems related to the part build interact with. Although desktop vat photopolymerization machines may pull parts up from the resin, industrial-sized machines typically pull parts down into the resin, which requires a UV light source above the vat curing each layer as a build platform incrementally lowers. As each layer cures, a recoater uses a blade to smooth a new coat of liquid resin over the previous layer. Resin delivery systems in an industrial vat photopolymerization machine can hold over a hundred gallons of material.

During setup, it is important to confirm that the resin being used was stored in securely fastened light-resistant containers, which shield it from sunlight. Any resin material exposed to light may partially cure, which can adversely affect the quality of the print. The resin storage area should also be temperature-controlled to between 60–80°F (16–27°C). Temperatures outside that range may cause the resin to prematurely cure, making it unusable. Maintaining proper environmental control in storage and in the machine is especially critical to maintaining quality and repeatability when blending older resin and newer resin over time.

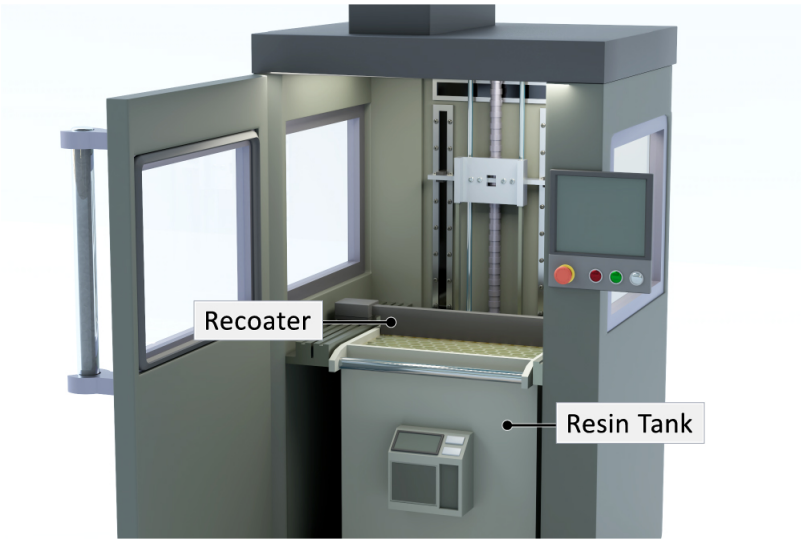

Figure 1:

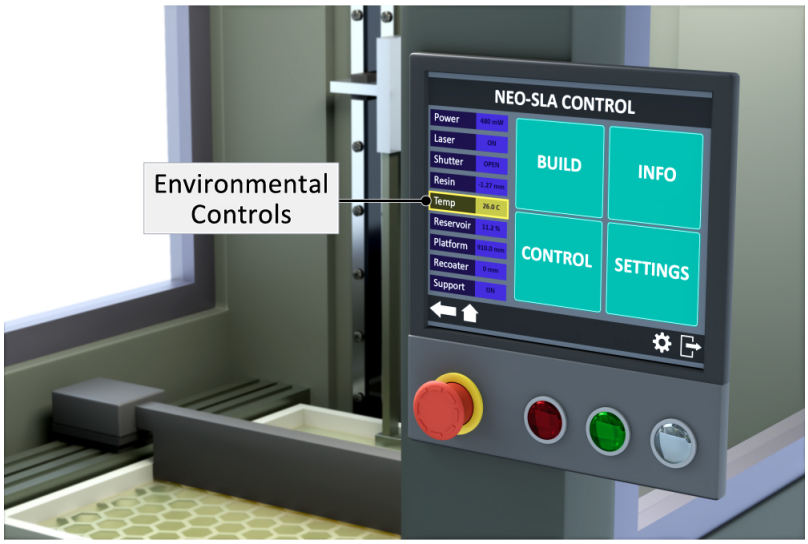

Figure 2: An operator can monitor environmental controls through the control panel.

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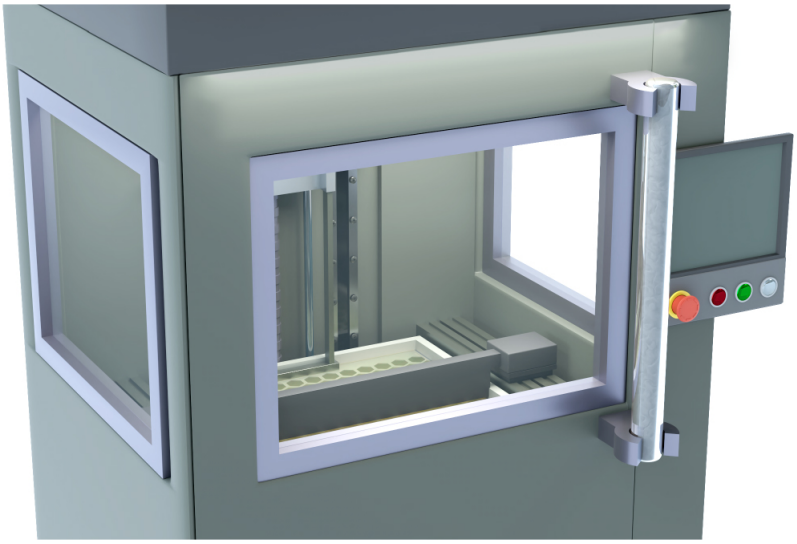
Machine Components

Many industrial vat photopolymerization resin tanks are self-contained and interchangeable. Typically, each vat will have a label with information about the material in the container, which helps prevent accidental usage of incorrect or expired material during setup. Interchangeable resin tanks enable an operator to replace an old vat with a new one without coming into contact with any build chamber components. It can also quicken the setup process by enabling an operator to warm the resin to the optimal build temperature.



The build chamber, or print chamber, confines the laser beam and blocks any UV radiation from the laser from escaping the build area. The build chamber also helps maintain optimal build temperature. Setup tasks for vat photopolymerization will typically involve pre-heating the build chamber to the ideal temperature for the resin material being used.

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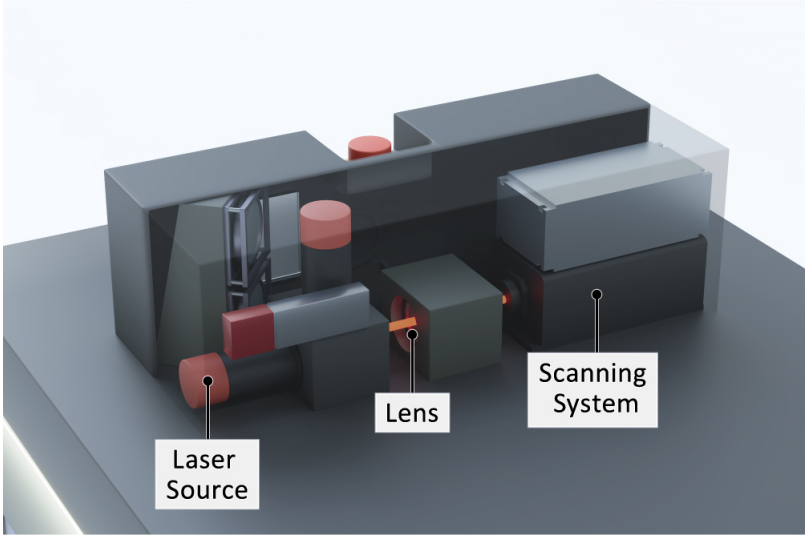

The laser of a vat photopolymerization machine is located near the top of the build chamber behind a protective window. When used under normal operating conditions, lasers require no special safety precautions. During setup, vat photopolymerization machines should have a safety interlock to protect the operator from accidental laser exposure. If an operator uses the machine without the interlocks being engaged or operating properly, eye damage can occur from viewing the beam directly or any kind of reflection.

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Vat Calibration

Common setup tasks during vat photopolymerization involve calibrating the operation of the resin tank, including the temperature needed for the resin.

If using an interchangeable vat, typically, the recoater blade should stay with the vat when changing material to avoid material cross-contamination. When installing the vat, an operator should wear personal protective equipment (PPE), like gloves, to avoid exposure to the resin material. When replacing a recoater, an operator will need to calibrate its operation, which typically involves running an automated sequence on the machine. An operator must also calibrate the elevator on the vat, which raises and lowers the build platform at specific intervals.

Proper setup helps operators maintain a controlled and consistent temperature during the build process, which is critical to industrial vat photopolymerization and repeatability. Warming the material before the print may take a half-hour or more. An operator will typically want to maintain a vat temperature warm enough to reduce the viscosity, or thickness, of the liquid resin. However, liquid resin warmed too long or at too high a temperature may cause a finished part to be too fragile or soft.

Figure 1: An operator should wear PPE to avoid exposure to the resin material. Image courtesy of Stratasys.

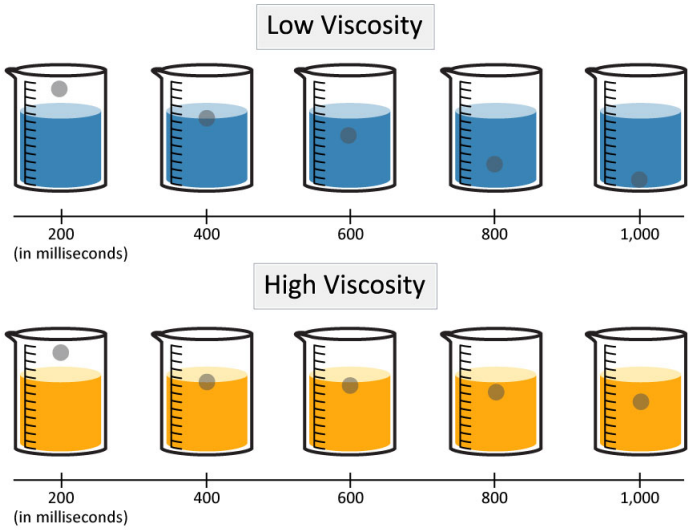

Figure 2: Viscosity can be measured by how quickly an object of known mass sinks into a liquid.

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Vat Photopolymerization Component and System Review

1. Typically, desktop vat photopolymerization machines are used for end-use part production. **FALSE**
2. Vat photopolymerization machines usually have their own proprietary software that will take STL file information and translate it into slices, or layers, that the machine will use to build the part. **TRUE**
3. Most industrial-size vat photopolymerization machines build objects from the top down. **FALSE**
4. All vat photopolymerization machines consist of a resin tank, build chamber, and laser. **TRUE**
5. An operator will typically want to maintain a vat temperature cool enough to increase the viscosity of the liquid resin. **FALSE**

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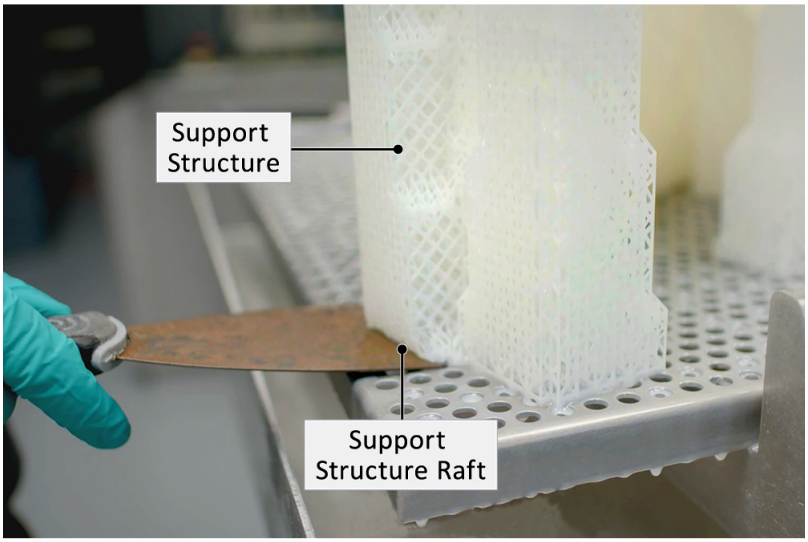
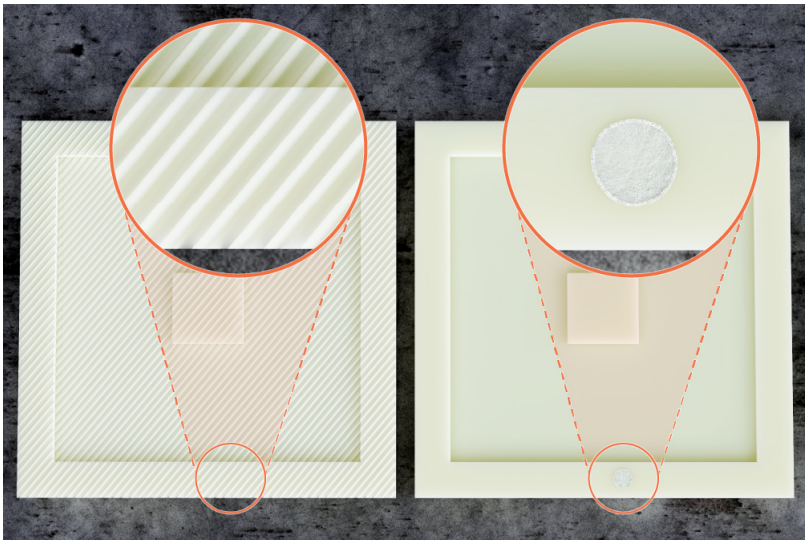
Build Orientation



In most cases, support structures are needed to create reliable, repeatable results in vat photopolymerization. How those support structures are configured during setup can affect the quality of the print. For example, to avoid peel force damage when removing a part from the build platform, an operator will use a support structure raft, or base.



A vat photopolymerization part build may also require a specific surface texture, or roughness, which can help hide blemishes caused by removing support structures or peeling the part from the build platform.

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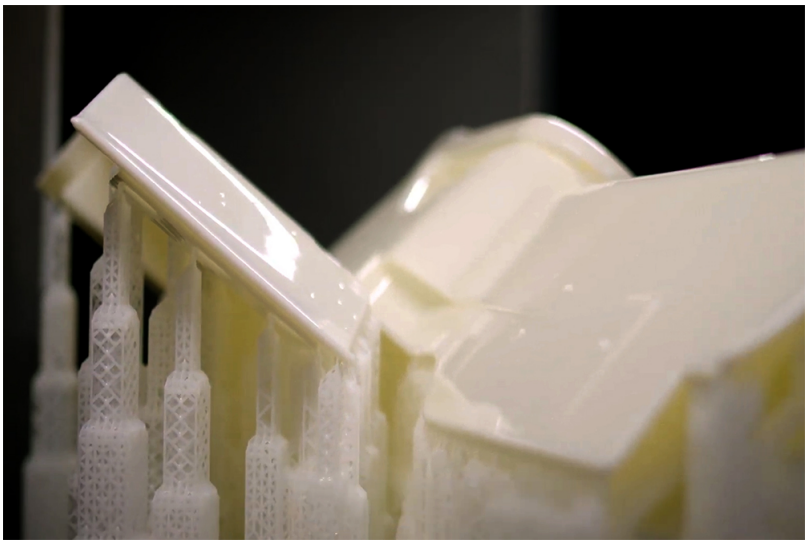
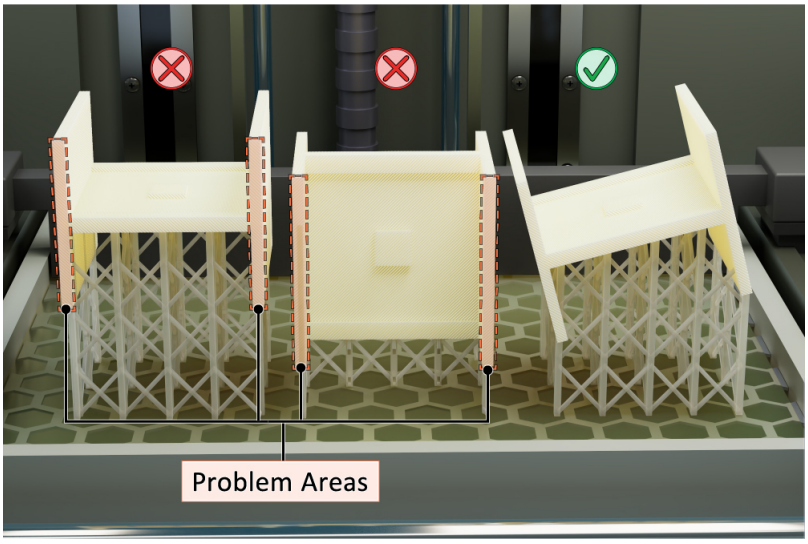


Vat photopolymerization machines can create multiple parts in one build batch. An operator can orient those parts before a build in many different combinations. For example, software can stack sets of parts separately or together during a build.



Differential shrink can occur during a vat photopolymerization build when a large area of a part rapidly transitions to a smaller area. Operators can compensate for differential shrink by rotating the part a certain way or adjusting support structure settings during the part build.

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Build Surface Setup

Before beginning a vat photopolymerization print, an operator must set up the build platform in the start position, which is considered the best position to start a successful print.

Any new build platform, once set in the start position, requires leveling to make it flat, or parallel, to the base of the vat and the same height as the surface of the liquid resin. When leveling a print platform to the start position, an operator must avoid partially submerging the build platform in the resin. However, the operator must also avoid setting the platform level above the resin surface.

Bubbles in the vat may occur, which an operator must remove before the print begins to avoid print errors. For example, bubbles in the resin might affect the adhesion between different layers in the final part. The process for removing bubbles is typically automated, though it may vary from machine to machine.

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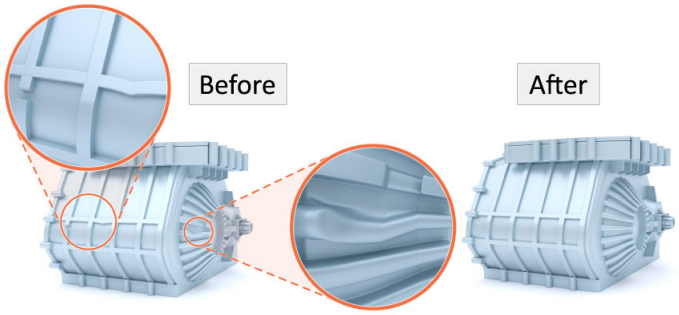
Laser Calibration



The power of the laser will decay over time and with usage, so operators should examine laser performance during setup at regular intervals depending on how often the machine is used. If there are visible changes in print quality, an operator may need to run the automated calibration procedure multiple times. In addition, an operator can print a calibration model to test the performance of the laser. Calibration models are specially designed programs that test the laser settings to determine if they need adjustment. For example, a calibration model can test the minimum width and height precision of the laser. If the laser creates too wide of a path, the operator may need to increase the laser exposure. A calibration model can also test the viscosity of the resin by examining how effectively the material drains out of tight, complex spaces that the laser creates. If unable to drain effectively, trapped resin can lead to loss of elaborate detailing.

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Setup Review

- **Build File Orientation**: An operator ensures that this matches the job file and is compatible with the digital slicer.
- **Laser**: The power of this will decay over time in a vat photopolymerization machine.
- **Bubbles**: An operator removes these before the print begins to avoid print errors.
- **Calibration Model**: A specially designed program that tests the laser settings to determine if they need adjustment.

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Class Vocabulary

Additive Manufacturing AM. The process of joining or solidifying materials to make an object based on a three-dimensional (3D) computer model. Additive manufacturing methods typically build up layers of material to create an object.

AM Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional (3D) computer model. AM methods typically build up layers of material to create an object.

Build Chamber The area contained within an additive manufacturing (AM) machine where the part is actually built. Many build chambers, also called print chambers, are enclosed to improve the safety of the manufacturing operation.

Build File Computer information that provides detailed instructions to an additive manufacturing machine. Build files include information about the exact shape and size of each part layer.

Build Platform The flat surface on which a part is additively manufactured. The build platform is a permanent surface from which parts are removed.

Calibration Model The comparison and adjustment of a small model built with unknown accuracy to a model with a known, accurate standard. Calibration models in vat photopolymerization are built with a designed program that tests the laser settings to determine if they need adjustment.

Computer-Aided Design CAD. A computer software program that aids in the automated design and technical precision drawing of a part, product, process, or building. Computer-aided design can create three-dimensional digital models used for additive manufacturing.

Cure To cause a material to bond and solidify by permanently cross-linking its molecules. Curing techniques for photopolymers include exposing the material to ultraviolet light during the production process.

Differential Shrink Reduction in size in a joint area of a part build. Differential shrink is a result of warping or reduction in size occurring in a joint area of a part build.

End-Use Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.

Environmental Control Process designed to control the temperature and gases in an additive manufacturing operation. Environmental controls maintain things like temperature, which helps to ensure that all materials for all builds are of the same quality or properties.

G Code A method of programming that pairs address letters with numerical values to form words. G code programs are used in additive manufacturing, CNC machining, and hybrid machining.

Laser An intense beam of light that can be precisely aimed and controlled. Lasers are sometimes used to selectively solidify or fuse materials in additive manufacturing methods, including directed energy deposition, powder bed fusion, and vat photopolymerization.

Leveling Keeping the surface of one or more components completely horizontal and in alignment. Leveling the build platform in vat photopolymerization ensures that a part is constructed accurately.

Measured Spot The actual dimensions of a part. Measured spots outside of part specifications are typically discarded or reworked.

Peel Force A material's ability to resist forces. The peel force needed for vat photopolymerization parts is typically low.

Personal Protective Equipment PPE. Any clothing or device used to minimize exposure to hazards and prevent injuries. Personal protective equipment may include safety glasses, hard hats, and steel-toed shoes.

Photopolymers A type of fluid plastic that cures and hardens when exposed to light. Photopolymers, or photoreactive resins, appear to be liquids but are technically viscous solids and are used in material jetting and vat photopolymerization.

Photoreactive Resins A material that chemically reacts to exposure to ultraviolet (UV) light. Photoreactive resins solidify and harden when exposed to UV light.

Print Chamber The area contained within an additive manufacturing (AM) machine where the part is actually built. Many print chambers, also called build chambers, are enclosed to improve the safety of the manufacturing

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operation.

Recoater A device that slides forwards and backwards over the surface of photopolymer material held in a vat. Recoaters are used by vat photopolymerization) systems to smooth photopolymer material layers over a build platform.

Repeatability The ability of a machine or process to continually deliver accurate and precise results. Repeatability measures how consistently a machine can replicate identical or near-identical parts.

Resin Resins that are photopolymer will harden when exposed to light and are used in material jetting and vat photo polymerization.

Safety Interlock A safety device that disables or prevents a machine startup if a guard or door remains open. Safety locks can protect the operator from accidental laser exposure.

Scrap Unusable material produced during a manufacturing process. Scrap is a waste product of manufacturing that can increase overall production costs.

Setup All the necessary preparation of tooling and fixturing that occurs on a machine before operation begins. Proper machine setup is an important factor in optimizing tool life.

Software The programs, instructions, and formulas that structure the actions of a computer. Software controls computer hardware functions and operations.

Start Position Where the operator places the build platform in a resin tank during vat photopolymerization. The start position is considered the best position for a successful vat photopolymerization print.

STL Computer files that represent 3D models as a series of interconnected triangles. STL files are converted from CAD files and are used in additive manufacturing machines.

Support Structure Raft The section of the part that makes contact, and is usually secured, to the build platform. The support structure raft serves as a foundation on which the rest of the components are built.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Target Spot The desired dimensions of a part. Target spots represent the spefications of the designer.

Test Print A procedure performed during the additive manufacturing process to assess the performance of the machine. Test prints are often used to calibrate the accuracy of the UV laser in vat photopolymerization.

Three-Dimensional 3D. Having height, width, and depth. Three-dimensional parts are created during additive manufacturing (AM) processes.

Ultraviolet UV. A potentially harmful wavelength of light that is below, or shorter than, violet on the light spectrum. Ultraviolet light is used to selectively harden a photopolymer in vat photopolymerization processes.

UV Ultraviolet. A potentially harmful wavelength of light that is below, or shorter than, violet on the light spectrum. UV light is used to selectively harden a photopolymer in vat photopolymerization processes.

Vat An industrial receptacle used to hold fluids. A vat is used to hold liquid photopolymers in vat photopolymerization processes.

Vat Photopolymerization An additive manufacturing (AM) method that builds a part by curing layers of photopolymer with light. Vat photopolymerization processes include stereolithography (SLA) and digital light processing (DLP).

Viscosity A fluid's resistance to flow. Viscosity in vat photopolymerization generally means that the hotter a resin is, the lower its viscosity.

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